

MODULE - IV

Simple Stress & Strain

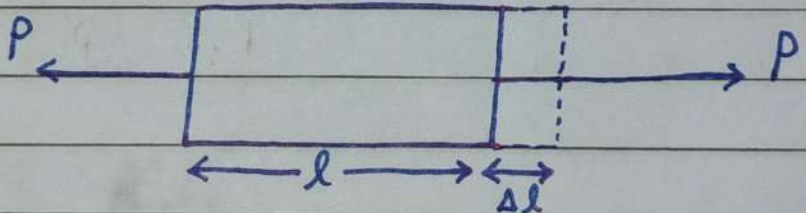
Stress

Stress at any point is defined as internal resistance force per unit area. It is denoted by σ .

$$\sigma = \frac{P}{A} \text{ N/m}^2 (\text{or Pascal}).$$

Strain

Strain is defined as the ratio of change in length to the original length. It is denoted by ϵ .

$$\epsilon = \frac{\Delta l}{l}$$


The diagram shows a rectangular bar of original length l . Two horizontal arrows labeled P point outwards from the left and right faces of the bar, representing tensile forces. Below the bar, a double-headed arrow indicates the original length l . To the right of the bar, a dashed vertical line indicates the new position of the right end after extension. A double-headed arrow between the original right edge and the dashed line is labeled Δl , representing the change in length.

Hooke's Law

According to Hooke's Law, stress is directly proportional to strain, up to elastic limit.

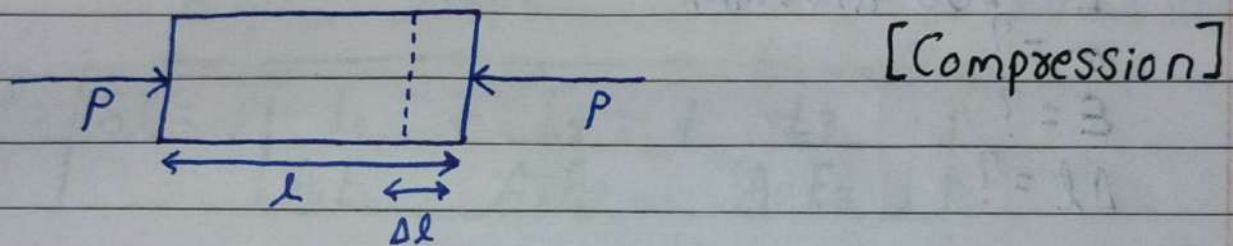
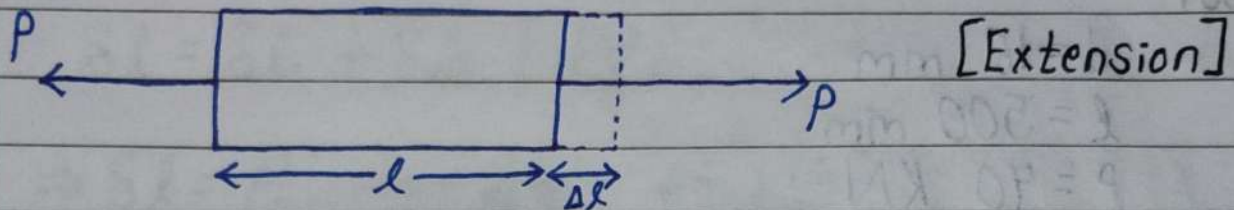
$$\boxed{\sigma \propto \epsilon}$$

$$\Rightarrow \sigma = E \epsilon$$

$$\text{i.e. } E = \frac{\sigma}{\epsilon}$$

$E \rightarrow$ Young's Modulus (Modulus of Elasticity).

Extension/Compression of Bar



$$E = \frac{\sigma}{\epsilon}$$

$$\Rightarrow E = \frac{P}{A} \cdot \frac{l}{\Delta l} \quad \left[\sigma = \frac{P}{A} \text{ \& } \epsilon = \frac{\Delta l}{l} \right]$$

$$\Rightarrow \Delta l = \frac{Pl}{EA}$$

Q. A circular rod of diameter 16 mm and 500 mm long is subjected to a tensile force 40 kN. The Young's Modulus of Elasticity of steel is taken 200 kN/mm². Find stress, strain and elongation to applied load?

Solⁿ:

$$d = 16 \text{ mm}$$

$$l = 500 \text{ mm}$$

$$P = 40 \text{ kN}$$

$$E = 200 \text{ kN/mm}^2$$

$$\sigma = ?$$

$$\epsilon = ?$$

$$\Delta l = ?$$

We know,

$$\text{Area of the rod, } A = \frac{\pi d^2}{4} = 201.06 \text{ mm}^2$$

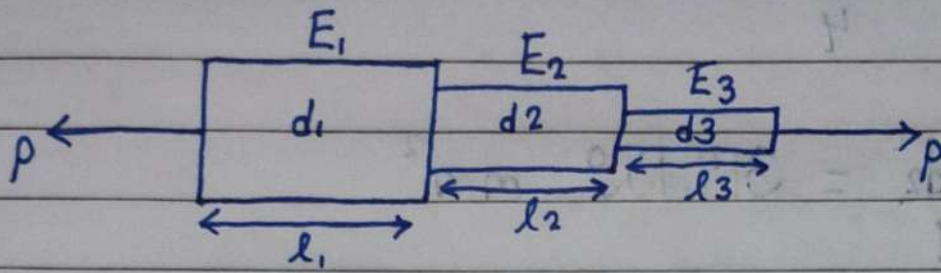
$$\sigma = \frac{P}{A} = 0.1989 \text{ kN/mm}^2$$

$$\epsilon = \frac{\Delta l}{l}$$

$$\Delta l = \frac{Pl}{EA} = \frac{40 \times 500}{200 \times 201.06} = 0.4973 \text{ mm}$$

$$\epsilon = \frac{\Delta l}{l} = \frac{0.4973}{500} = 0.000994$$

Bars with Varying Cross-section in Steps

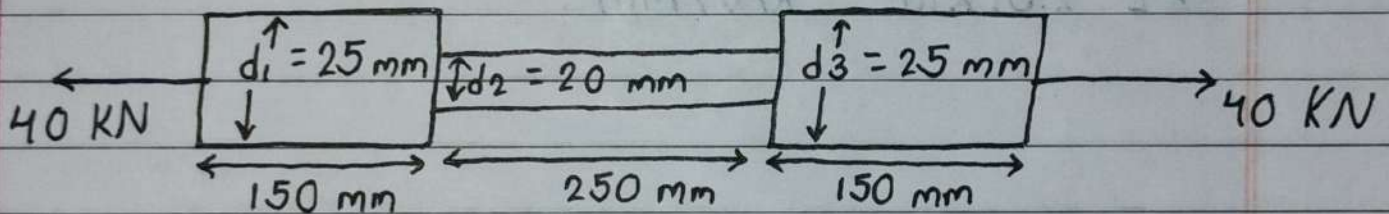


$$\delta l = \delta l_1 + \delta l_2 + \delta l_3$$

$$\Rightarrow \delta l = \frac{PL_1}{A_1 E_1} + \frac{PL_2}{A_2 E_2} + \frac{PL_3}{A_3 E_3}$$

$$\Rightarrow \delta l = P \left[\frac{L_1}{A_1 E_1} + \frac{L_2}{A_2 E_2} + \frac{L_3}{A_3 E_3} \right]$$

Q.



The bar shown in figure is tested in UTM. It is observed that at a load of 40 kN , the total extension of the bar is 0.28 mm . Determine the Young's Modulus?

Solⁿ:

$$\Delta l = 0.28 \text{ mm}$$

$$P = 40 \text{ kN}$$

$$d_1 = 25 \text{ mm}, l_1 = 150 \text{ mm}$$

$$d_2 = 20 \text{ mm}, l_2 = 250 \text{ mm}$$

$$d_3 = 25 \text{ mm}, l_3 = 150 \text{ mm}$$

Now,

$$A_1 = A_3 = \frac{\pi d_1^2}{4} = 490.873 \text{ mm}^2$$

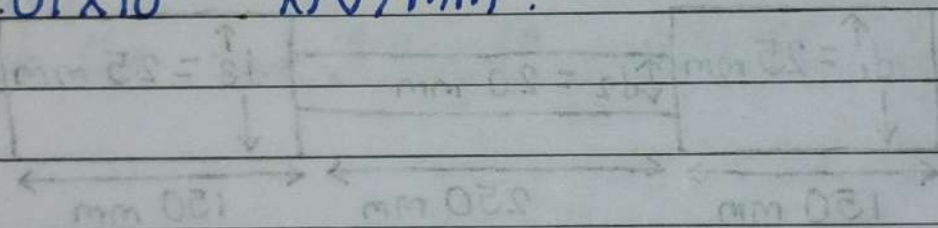
$$A_2 = \frac{\pi d_2^2}{4} = 314.159 \text{ mm}^2$$

We know,

$$\Delta l = \frac{P}{E} \left[\frac{l_1}{A_1} + \frac{l_2}{A_2} + \frac{l_3}{A_3} \right]$$

$$\therefore E = \frac{40}{0.28} \left[\frac{150}{490.873} + \frac{250}{314.159} + \frac{150}{490.873} \right]$$

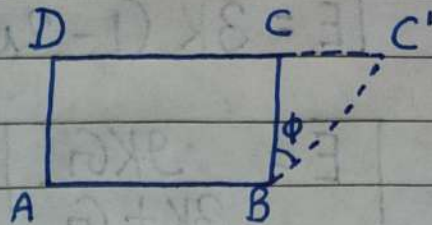
$$\Rightarrow E = 2.01 \times 10^3 \text{ KN/mm}^2.$$



Shear Stress

A shear stress is defined as a ratio of shearing force to the unit area.

$$\tau = \frac{P}{A}$$

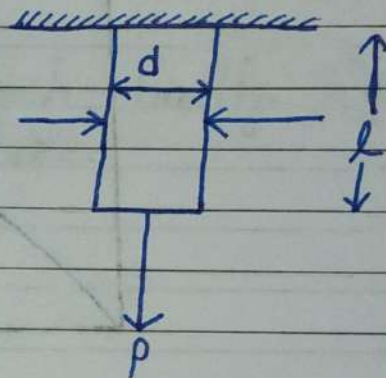


Poisson's Ratio

It is defined as a ratio of lateral strain to the longitudinal strain.

$$\mu = \frac{\text{Lateral Strain}}{\text{Longitudinal Strain}}$$

$$\mu = - \frac{\Delta d/d}{\Delta l/l}$$



Elastic Constants

E = Modulus of Elasticity (Young's Modulus)

G = Modulus of Rigidity

K = Bulk Modulus of Elasticity

$$E = \frac{\sigma}{\epsilon}, \quad G = \frac{\tau}{\phi} \quad \& \quad K = \frac{\sigma_n}{\epsilon_v} = \frac{\sigma_n}{(\epsilon_x + \epsilon_y + \epsilon_z)}$$

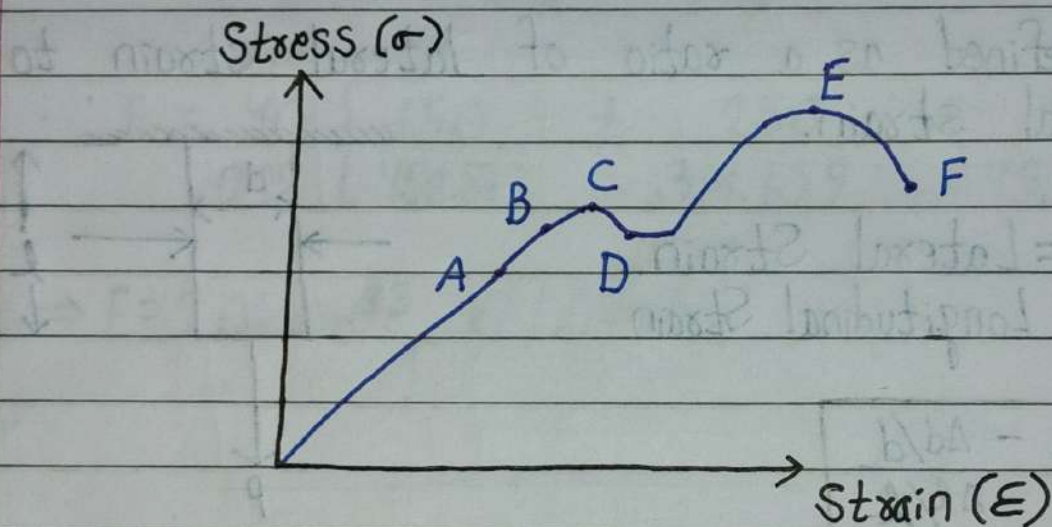
Relationship Between Elastic Constants

$$E = 2G(1 + \mu)$$

$$E = 3K(1 - 2\mu)$$

$$E = \frac{9KG}{3K + G}$$

Stress - Strain Diagram



i. Proportional Limit (A):

It is a limiting value of the stress upto which stress is proportional to strain.

ii. Elastic Limit (B):

This is the limiting value of the stress upto which if the material is stretched and then released, strain disappears completely and original length of material is regained. It is slightly beyond to the proportional limit.

iii. Upper Yield Point (C):

This is the stress at which the load starts reducing and ~~extent~~ extension increase.

iv. Lower Yield Point (D):

At this stage the stress remain same but strain increases for some time.

v. Ultimate Stress (E):

This is the maximum stress, the material can resist this is called neck-formation.

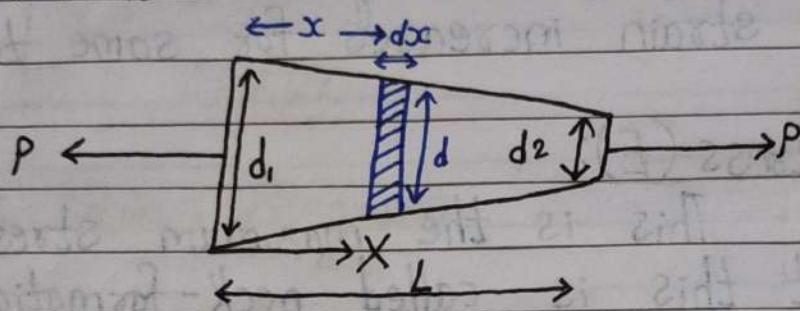
vi. Breaking Point (F):

The stress at which finally the specimen breaks / fail. This is called fracture.

V. Imp
Q.

A tapering rod has a diameter d_1 at one end and it tapers uniformly to a diameter d_2 at other end, in a length ~~capital~~ L as shown in figure. If modulus of elasticity of the material is E , find its change in length when subjected to axial load P ?

Solⁿ:



The rod is uniform so,

$$\frac{d_1 - d_2}{L} = \frac{d_1 - d}{x}$$

$$\Rightarrow d = d_1 - \frac{x(d_1 - d_2)}{L}$$

$$\Rightarrow d = d_1 - Kx \quad \left[\frac{d_1 - d_2}{L} = \text{constant} \rightarrow K(\text{say}) \right]$$

Now, the change in length of the strip is,

$$\Delta l_s = \frac{P \cdot dx}{A_s \cdot E}$$

$$\text{Area of the strip, } A_s = \frac{\pi d^2}{4} = \frac{\pi}{4} (d_1 - Kx)^2$$

$$\text{So, } \Delta l_s = \frac{4P \cdot dx}{\pi (d_1 - kx)^2 \cdot E}$$

$$\int_0^L \Delta l_s = \int_0^L \frac{4P dx}{\pi (d_1 - kx)^2 \cdot E}$$

$$\Rightarrow \Delta l = \frac{4P}{\pi E} \int_0^L \frac{dx}{(d_1 - kx)^2}$$

$$\Rightarrow \Delta l = \frac{4P}{\pi E} \int_0^L \frac{1}{(d_1 - kx)^2} \cdot dx$$

$$\text{Let, } t = d_1 - kx$$

$$\Rightarrow \frac{dt}{dx} = -k$$

$$\Rightarrow dx = \frac{-dt}{k}$$

$$\Delta l = \frac{4P}{\pi E} \int_0^L \frac{1}{t^2} \cdot \left(\frac{-dt}{k} \right)$$

$$= \frac{-4P}{\pi k E} \int_0^L \frac{1}{t^2} \cdot dt$$

$$= \frac{-4P}{\pi k E} \left[\frac{-1}{t} \right]_0^L$$

$$= \frac{4P}{\pi k E} \left[\frac{1}{d_1 - kx} \right]_0^L$$

$$\Delta l = \frac{4P}{\pi KE} \left[\frac{1}{d_1 - KL} - \frac{1}{d_1} \right]$$

$$= \frac{4P}{\pi KE} \left[\frac{d_1 - d_1 + KL}{d_1(d_1 - KL)} \right]$$

$$= \frac{4P}{\pi KE} \left[\frac{KL}{d_1(d_1 - KL)} \right]$$

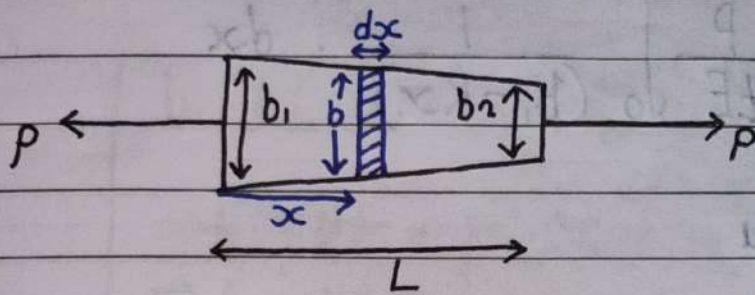
$$= \frac{4P}{\pi E} \times \frac{L}{(d_1 - d_2)} \times \frac{KL}{d_1(d_1 - KL)} \quad \left[K = \frac{d_1 - d_2}{L} \right]$$

$$= \frac{4P}{\pi E} \times \frac{L}{\cancel{(d_1 - d_2)}} \times \frac{\cancel{L}}{d_1 d_2} \times \frac{\cancel{(d_1 - d_2)}}{\cancel{L}} \quad \left[\begin{array}{l} d_2 = d_1 - KL \\ K = (d_1 - d_2)/L \end{array} \right]$$

$$= \frac{4PL}{\pi E d_1 d_2}$$

$$\therefore \Delta l = \frac{4PL}{\pi E d_1 d_2}$$

Q. A bar of uniform thickness t tapers uniformly from a width b_1 at one end to b_2 at the other end in a length L as shown in figure. Find the expression for the change in length of the bar when subjected to an axial load P .



Solⁿ:

The bar is uniform, so

$$\frac{b_1 - b_2}{L} = \frac{b_1 - b}{x}$$

$$\Rightarrow x \left(\frac{b_1 - b_2}{L} \right) = b_1 - b$$

$$\Rightarrow b = b_1 - \left(\frac{b_1 - b_2}{L} \right) x$$

$$\Rightarrow b = b_1 - Kx \quad \left[\because K = \frac{b_1 - b_2}{L} \right]$$

Area of the strip, $A_s = bt = (b_1 - Kx)t$

Now, the change in length of the strip is,

$$\Delta l_s = \frac{P \cdot dx}{A_s \cdot E}$$

$$\Rightarrow \Delta l_s = \frac{P \cdot dx}{(b_1 - Kx) t E}$$

$$\Rightarrow \int_0^L \Delta l_s = \int_0^L \frac{P}{(b_1 - Kx) t E} \cdot dx$$

$$\Rightarrow (\Delta l)_{\text{bar}} = \frac{P}{tE} \int_0^L \frac{1}{(b_1 - Kx)} \cdot dx$$

$$\text{Let, } b_1 - Kx = u$$

$$\Rightarrow -K = \frac{du}{dx}$$

$$\Rightarrow dx = -\frac{du}{K}$$

$$(\Delta l)_{\text{bar}} = \frac{P}{tE} \int_0^L \frac{1}{u} \cdot \left(-\frac{du}{K} \right)$$

$$= -\frac{P}{tKE} [\log u]_0^L$$

$$= -\frac{P}{tKE} [\log(b_1 - KL) - \log(b_1)]$$

$$= -\frac{P}{tKE} [\log(b_1 - KL) - \log(b_1)]$$

$$= \frac{PL}{tE(b_1 - b_2)} [\log b_1 - \log(b_1 - KL)]$$

$$\left[\because K = \frac{b_1 - b_2}{L} \right]$$

$$= \frac{PL}{Et(b_1 - b_2)} \log \left(\frac{b_1}{b_1 - KL} \right)$$

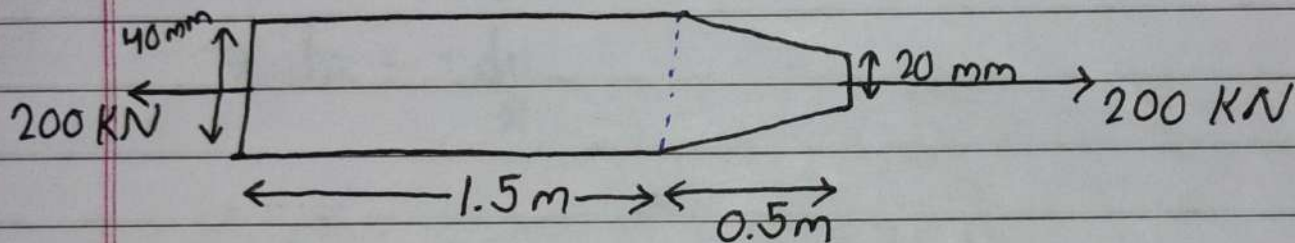
$$\therefore (\Delta l)_{\text{bar}} = \frac{PL}{Et(b_1 - b_2)} \log \frac{b_1}{b_2}$$

$$\left[\begin{aligned} b_1 - KL &= b_1 - \left(\frac{b_1 - b_2}{1} \right) \leftarrow \\ \Rightarrow b_1 - KL &= \cancel{b_1} - \cancel{b_1} + b_2 \\ \Rightarrow b_1 - KL &= b_2 \end{aligned} \right]$$

Numericals

Q. A two meter long steel bar is having uniform diameter of 40 mm for a length of ~~1.5~~ 1.5 m and in the next 0.5 m its diameter gradually reduces from 40 mm to 20 mm as shown in figure. Determine the elongation of this bar when subjected to an axial tensile load of 200 kN? $[E = 200 \text{ GN/m}^2]$

Solⁿ:



$$d_1 = 40 \text{ mm} = 40 \times 10^{-3} \text{ m}$$

$$d_2 = 20 \text{ mm} = 20 \times 10^{-3} \text{ m}$$

$$E = 200 \text{ GN/m}^2 = 200 \times 10^9 \text{ N/m}^2$$

$$P = 200 \text{ kN}$$

$$L_1 = 1.5 \text{ m}$$

$$L_2 = 0.5 \text{ m}$$

$$A_1 = \frac{\pi d_1^2}{4} = 0.0012566 \text{ m}^2$$

$$\therefore \delta l = \delta l_1 + \delta l_2 = \frac{PL_1}{A_1 E} + \frac{4PL_2}{\pi E d_1 d_2}$$

$$\Rightarrow \delta l = 0.001989 \text{ m} = 1.98 \text{ mm}$$

Q. A bar of 25 mm diameter is tested in tension. It is observed that when a load of 60 kN is applied, the extension measured over a gauge length of 200 mm is 0.12 mm and contraction in diameter is 0.0045 mm. Find poisson's ratio and elastic constant?

Solⁿ:

$$P = 60 \text{ kN} = 60 \times 10^3 \text{ N}$$

$$l = L = 200 \text{ mm} = 200 \times 10^{-3} \text{ m}$$

$$\Delta l = 0.12 \text{ mm} = 0.12 \times 10^{-3} \text{ m}$$

$$d = 25 \text{ mm} = 25 \times 10^{-3} \text{ m}$$

$$\Delta d = 0.0045 \text{ mm} = 0.0045 \times 10^{-3} \text{ m}$$

Now,

$$A = \frac{\pi d^2}{4} = 0.0004908 \text{ m}^2$$

$$\therefore \mu = -\left(\frac{\Delta d/d}{\Delta l/l}\right) = -\left(\frac{0.0045/25}{0.12/200}\right) = -0.3 = 0.3$$

$$E = \frac{PL}{A \Delta l} = \frac{60 \times 10^3 \times 200 \times 10^{-3}}{0.0004908 \times 0.12 \times 10^{-3}} = 203.74 \text{ GN/m}^2$$

$$\Rightarrow E = 203.74 \times 10^9 \text{ N/m}^2$$

$$E = 2G(1 + \mu)$$

$$\Rightarrow G = \frac{E}{2(1 + \mu)} = \frac{203.74 \times 10^9}{2(1 + 0.3)} = 78.36 \times 10^9 \text{ N/m}^2$$

$$k = \frac{E}{3(1 - 2\mu)} = 169.78 \times 10^9 \text{ N/m}^2$$